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Reliance Jio — Subscriber Retention and Revenue Intelligence

Assessment of churn-driven revenue leakage across the Indian wireless base, and a proposed analytics and machine-learning programme to address it.

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Independent third-party assessment prepared for internal engagement use. Not a record of, prepared on behalf of, or endorsed by the subject company. Quantitative indicators are as-disclosed and unaudited; estimates are marked directional where internal data is unavailable. Basis of preparation and notes at the end of this document.

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Basis of preparation

Revision history

Version	Date	Author	Change	Reviewed
v0.1	14 Aug 2026	A. Ranjan	Initial sizing, KPI pull	—
v0.2	02 Sep 2026	A. Ranjan	Added uplift rationale; reworked section 4 after review	S.K.
v0.3	17 Sep 2026	A. Ranjan	Added open items; phasing owners still to be confirmed	—

Distribution: engagement team; practice lead. Not for onward circulation.

Status: draft. Section 10 phasing is unowned and section 11 remains open.

1. Summary of view

Jio's headline numbers look healthy. The base has crossed 515 million, ARPU has climbed for four consecutive quarters, and data traffic is up 34% year on year.¹ None of that is the problem.

The problem sits underneath it. Monthly churn has held at 1.8–1.9% for a year with no sign of compression. At the current base that is roughly **9.3 million subscribers walking out every month**, and on reported ARPU it puts something in the order of **₹23,800 crore of annual revenue in play** — call it 16% of the top line.

What makes this worth a programme rather than a campaign is that the Indian market has run out of new people. The wireless base is saturated at about 1.17 billion connections. Industry portability requests have sat at 14–15 million a month since July 2025 and cumulative requests have passed 1.23 billion.² Growth is now almost entirely share transfer, which means retention economics have quietly become more important than acquisition economics, and most operators have not restructured around that.

Our working view is that the retention function is reactive by design. Subscribers get flagged once a port-out request lands, which is far too late. Offers are allocated on broad ARPU bands rather than individual propensity, so a meaningful share of retention spend goes to people who were never going to leave. And any question that needs CRM, billing, network and ticketing data in the same answer takes days of analyst time, so circle leadership is making decisions on stale numbers.

We propose three workstreams — diagnostics, predictive and prescriptive modelling, and a conversational analytics layer. Target is a 10% reduction in monthly churn, worth roughly **₹2,385 crore retained annually**. That target is deliberately conservative; see note 3 on why we have not modelled higher.

2. Where the base stands

Indicator	Q1 FY26	Q2 FY26	Q3 FY26	Q1 FY27
Closing subscriber base	498.1 M	506.4 M	515.3 M	533.0 M
Quarterly net additions	9.9 M	8.3 M	8.9 M	n/d
Monthly churn	1.8%	1.9%	1.8%	n/d
ARPU	₹203.3	₹211.4	₹213.7	₹215.6
Quarterly revenue	n/d	n/d	₹43,683 cr	n/d
Quarterly EBITDA	n/d	n/d	₹19,303 cr	n/d

Engagement over the same window:

Indicator	Q3 FY26
Total data traffic	62.3 bn GB, +34% YoY
Per-capita consumption	40.7 GB/month
Voice traffic	1.53 trn minutes
5G base	253 M, ≈53% of wireless traffic
Fixed broadband premises	25.3 M
AirFiber subscribers	11.5 M

Basis: company quarterly disclosures. n/d = not disclosed for that quarter.

The engagement picture is genuinely strong and improving. That matters for diagnosis: this is not a product-usage problem or a network-experience problem in aggregate. People are using the service heavily and leaving anyway. The causes are price sensitivity in the tail, service-assurance failure in specific journeys, and competitive offer response — all of which are individual-level effects that aggregate metrics hide.

3. Sizing the exposure

Monthly churn	=	1.8% × 515.3 M	≈	9.3 M exits / month
Monthly revenue risk	=	9.3 M × ₹213.7	≈	₹1,987 cr / month
Annualised			≈	₹23,800 cr
			≈	16% of annual revenue
At 10% churn reduction			≈	₹2,385 cr retained p.a.

Two caveats on this arithmetic. It applies reported ARPU uniformly, which overstates the loss because churn skews to the low-value tail — the true figure is lower, possibly materially. Against that, it ignores reacquisition cost, dealer commission and the SG&A drag of running a permanent win-back machine, which pushes the other way. We have not attempted to net these off without internal cost data. Treat ₹23,800 crore as an order-of-magnitude anchor, not a forecast.

4. What is driving it

4.1 Portability velocity, heavily concentrated

Sustained industry porting of 14–15 million requests a month is the clearest signal that competitive intensity has shifted entirely onto the installed base. It also carries a cost that does not show up in churn metrics: elevated dealer commissions and SG&A even in quarters where net adds stay positive.

Porting is not evenly spread.³

Circle cluster	Monthly porting requests
Uttar Pradesh East + West	3.3 M
Madhya Pradesh	1.4 M
Bihar	1.3 M

Three clusters account for a disproportionate share of national volume. They are the obvious pilot territory.

4.2 The low-ARPU tail is extremely price-elastic

After the last headline tariff revision the base contracted by 10.9 million in a single quarter, with losses concentrated almost entirely among low-value subscribers migrating to cheaper alternatives.

This is the most useful natural experiment available. One undifferentiated pricing action produced wildly differentiated churn outcomes, which is strong evidence that segment-level offer targeting needs to exist *before* the next revision rather than after it. If a future tariff move is on the roadmap, the modelling work in section 6 should be sequenced ahead of it.

4.3 Home connectivity provisioning is not keeping pace

AirFiber has scaled past 11.5 million subscribers. Provisioning throughput has not scaled with order intake, and the failure modes showing up in consumer grievance channels are consistent and specific:⁴

- advance payment taken, no engineer visit follows
- tickets raised against installation delay receive no callback, SMS or email
- relocation requests stay open across repeated follow-ups
- cancellation refunds remain pending after the order is abandoned

Each of these is operationally fixable and each has a direct line to both churn and regulatory exposure. We would treat this as a separate workstream rather than folding it into the wireless retention programme — the data, the owners and the interventions are all different.

4.4 UCC detection is now an obligation

The regulator's direction of 27 February 2026 requires all access providers to institutionalise AI/ML-based UCC detection, with inter-operator intelligence sharing.⁵ National spam volumes run at roughly 41.7 billion calls a year. Grievance-redressal rules are tightening in parallel.

Detection capability has moved from discretionary to mandatory. Worth noting mainly because it competes for the same data-science capacity as the retention programme.

5. What we could not establish

Listing this explicitly because several recommendations below are sensitive to it.

- **Churn denominator.** The company does not disclose whether reported monthly churn is computed on opening, closing or average base. We have assumed closing throughout. If it is average, our exposure figure moves by low single-digit percent.
- **Voluntary vs involuntary split.** No visibility into how much of the 1.8% is deliberate port-out versus dormancy, non-recharge or disconnection. These need completely different interventions, and the split materially changes programme design.
- **Circle-level churn.** Not separately disclosed. The circle ranking in 4.1 is inferred from portability concentration and is directional only.
- **Current retention offer economics.** We have no view of existing offer cost, redemption rates or control-group discipline. Uplift modelling in 6.3 assumes historical treatment and control assignment exists in campaign management. If it does not, phase 3 needs a randomised pilot first and the timeline extends.
- **Gross adds quality.** Net adds are visible, gross adds and the churn of recently acquired subscribers are not. Acquisition churning within 90 days would change where we focus.

None of these block starting. All of them need answering inside phase 1.

6. Proposed programme

6.1 Workstream 1 — diagnostics

Establish where churn concentrates and which levers actually move it, before building anything predictive.

Analysis	Question it settles
Circle and port-out heatmap	Which circles are losing base fastest, at what ARPU
Cohort and survival curves by joining month	Where tenure risk peaks; whether a 90-day cliff exists
RFM on recharge behaviour	How the prepaid base stratifies on recency, frequency, value
ARPU band × churn cross-tab	Quantifies the concentration in the low-value tail
Network KPI to churn correlation	Whether congested or low-SINR sites predict exits in the serving pin code
Recharge-gap analysis	How many silent days before a subscriber is effectively gone
5G migration funnel	Whether 5G adoption measurably lifts ARPU and suppresses churn
Ticket volume vs churn	Whether two or more unresolved tickets is a reliable early signal
Plan-migration flows	Where down-trading is happening after price changes

Output is an executive dashboard with circle, segment and tenure drill-downs, plus a written findings note. The findings note matters more than the dashboard; the point of this phase is to settle the open questions in section 5.

6.2 Workstream 2 — predictive modelling

The primary model predicts probability of exit or port-out at 30, 60 and 90 days. Logistic regression as a baseline, gradient-boosted trees as the likely champion.

The real difficulty is class imbalance. A 1.8% monthly event rate means a model that predicts "nobody churns" is 98.2% accurate and completely useless. Evaluation therefore runs on PR-AUC, recall at top decile and lift, and we would push back hard on any stakeholder request to report plain accuracy. Class weighting and threshold optimisation over resampling, on the basis that resampling distorts calibration and we need calibrated probabilities downstream for the offer economics.

SHAP attributions per subscriber feed the retention agent's screen and the conversational assistant. Without them the model is unusable in a contact-centre workflow — agents will not act on a score they cannot explain.

Alongside it, a CLV regression on expected 12-month contribution, so at-risk subscribers can be ranked by worth defending rather than by risk alone.

6.3 Workstream 2b — uplift

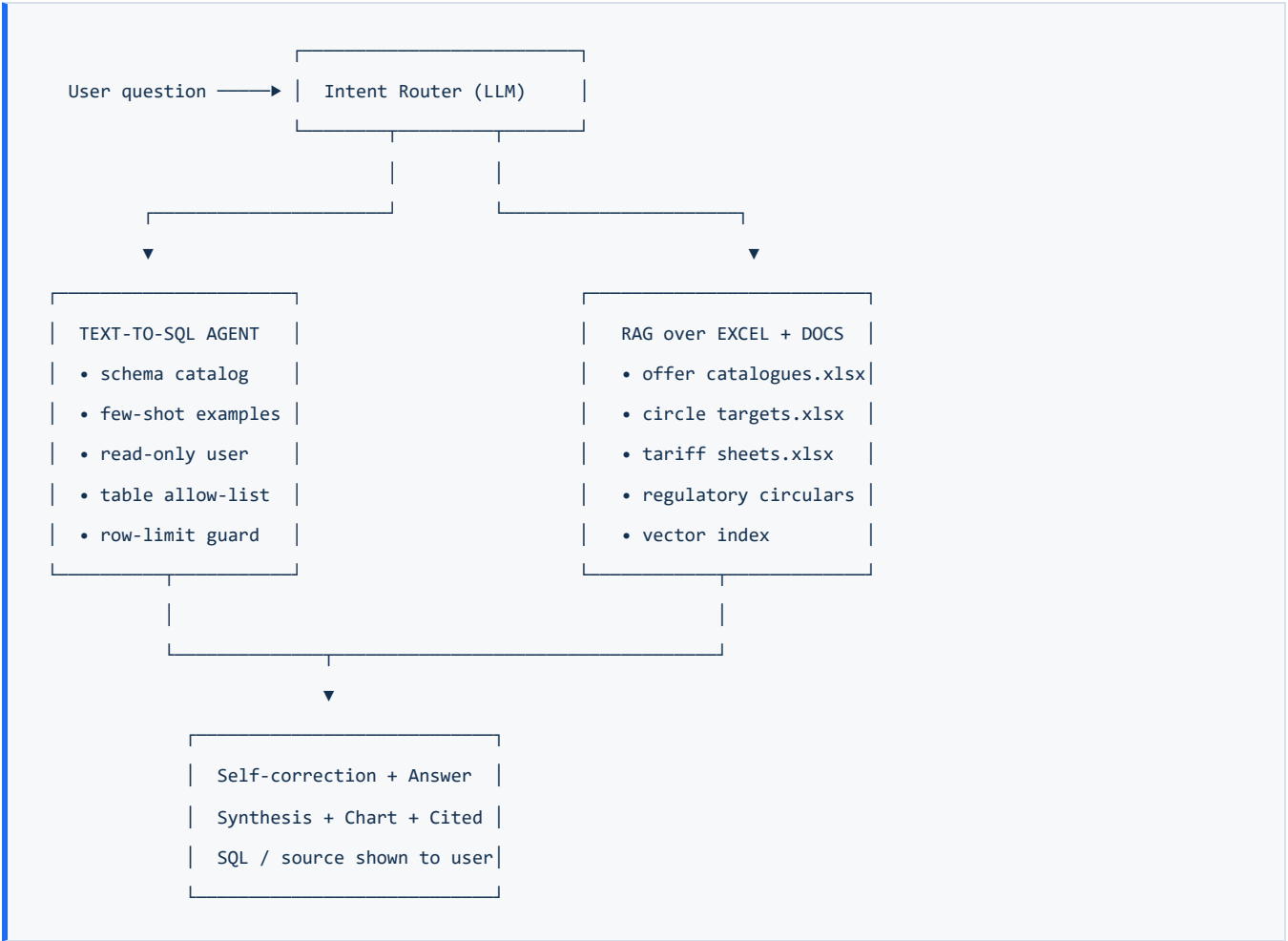
This is the piece that usually gets skipped and usually shouldn't.

A propensity model tells you who will leave. It does not tell you who will *stay because you intervened*. Those are different populations, and discounting the second group — people who were never going to leave — is the largest single source of wasted retention spend in conventional programmes. Two-model or causal-forest uplift estimation separates persuadables from sure-things and lost causes.

Dependent on historical treatment and control data existing. See section 5.

6.4 Workstream 3 — conversational analytics

The analyst bottleneck is a real constraint on everything above. A model that scores 515 million subscribers is worth very little if a circle head has to file a request and wait three days to ask what it says.



Queries it has to handle, and how each resolves:

Query	Path
"Top five circles by port-out last month, with average ARPU"	Warehouse SQL
"How many high-value subscribers are in the top risk decile this week"	SQL over model scores

Query	Path
"What retention offers are approved for Bihar this quarter"	Workbook retrieval
"Why is subscriber 98XXXXXXX high risk, and what should we offer"	SQL + SHAP + workbook
"Q3 actual churn against the retention target in the planning sheet"	SQL joined to workbook

That last row is the one that justifies the whole component. A meaningful share of commercial decisioning lives in spreadsheets outside the warehouse, so an assistant that only reads the database returns answers that are technically correct and commercially incomplete.

On correctness, the non-negotiables:

A semantic layer encoding certified definitions of churn, active subscriber, ARPU and net add, injected into context so the assistant cannot quietly apply a non-standard definition and produce a number that disagrees with the board pack. A curated few-shot query bank mapped to the production schema. SQL validation with an automatic correction loop on execution failure. Read-only service account, table allow-list, enforced row limits, hard block on DDL and DML — the assistant must not be able to modify data under any prompt, including a prompt embedded in a retrieved document. Prompt-injection filtering on retrieved content. Explicit abstention where confidence is low, in preference to a plausible wrong answer.

And a golden set of 50–100 certified question-answer pairs with execution accuracy tracked as a release gate. Without that there is no way to know whether a prompt change made things better or worse.

7. Adjacent opportunities

Both flagged rather than scoped. Each is a programme in its own right.

Home connectivity field operations. Addresses 4.3. Diagnostic layer on installation turnaround by pin code, ticket ageing, repeat truck-roll rate, order-to-activation drop-off. Predictive layer on SLA-breach classification per work order, cancellation and refund-request prediction, and engineer capacity forecasting by pin code. Conversational layer over the ticketing system and the roster and escalation workbooks.

UCC and fraud detection. Addresses 4.4, and is compulsory. Sender call-graph features (fan-out ratio, callee uniqueness, duration entropy), temporal burst detection, supervised UCC classification, unsupervised anomaly detection for SIM-box and revenue fraud, graph methods for organised rings.

8. Data requirements

System	What we need from it
Subscriber master / CRM	Demographics, tenure, circle, plan, device, KYC segment
Convergent billing and recharge	Recharge cadence, value, gaps, payment failures, plan history
CDR mediation	Voice, SMS and data session records for usage features
Portability gateway	Port-in / port-out request logs and outcome codes
Network OSS	Cell-site RSRP, SINR, PRB utilisation, drop-call rate, throughput
Care ticketing	Service requests, category, ageing, reopen rate, resolution
Field force management	Work orders, rosters, truck rolls, closure codes
Campaign management	Offer exposure, redemption, historical treatment/control assignment
App clickstream	Self-care usage, plan-browsing behaviour
Planning workbooks (Excel)	Circle targets, offer catalogues and margins, escalation matrices

The campaign management row is the dependency to chase first, because 6.3 falls over without it.

9. Architecture

Layer	Components
Storage	Warehouse on PostgreSQL; columnar store for feature tables
Processing	Python, pandas, PySpark at CDR scale
Analytics / BI	Power BI for executive views, Streamlit for analyst tooling
Modelling	scikit-learn, XGBoost, LightGBM, SHAP, lifelines, causal uplift libraries
Tracking	MLflow — registry, versioning, promotion gates
Generative AI	LLM-backed SQL agent with retrieval orchestration
Vector index	FAISS or managed equivalent for workbooks and circulars
Workbook ingestion	pandas + openpyxl, chunked and embedded on scheduled refresh
Delivery	Streamlit pilot, API-backed client at scale
Controls	Read-only role, table allow-list, SQL parser validation, row limits, injection filtering, full query audit log

10. Impact and phasing

Outcome	Basis
~₹2,385 cr retained annually	10% reduction against the ₹23,800 cr anchor
Reduced retention spend leakage	Uplift targeting stops discounting sure-things
Lower SG&A and commission drag	Fewer port-outs to win back
Shorter decision latency	Self-service replaces multi-day analyst cycles
Compliance	UCC detection per the 27 Feb 2026 direction

Phase	Scope	Owner	Target
1	Diagnostics across all circles; settle section 5 definitions	A. Ranjan	Q3 FY27
2	Churn and CLV models in shadow mode; validate lift on actuals	unassigned	TBC
3	Uplift model; controlled pilot in UP-E, UP-W, MP, Bihar	unassigned	TBC
4	Assistant pilot with circle leadership, gated on eval set	unassigned	TBC
5	National rollout; start adjacent workstreams	unassigned	TBC

Phasing beyond phase 1 is indicative. Owners and dates need confirming with the practice lead before this goes any further.

11. Open items

1. Confirm churn denominator with the engagement team — blocks final exposure sizing.
2. Establish whether historical treatment/control assignment exists in campaign management. Blocks 6.3.
3. Decide whether home connectivity (section 7) is in scope for this engagement or a separate proposal.
4. Phase 2–5 owners unassigned.
5. Reconcile the ₹23,800 cr anchor against a value-weighted churn estimate once segment-level ARPU is available. Current figure is knowingly conservative in one direction and aggressive in the other.

Basis of preparation

Prepared by A. Ranjan for internal engagement use. Quantitative indicators are drawn from the company's disclosed quarterly results, published regulatory performance indicators and industry trackers, as footnoted. Figures are as-disclosed and unaudited. Where internal data is unavailable, estimates are marked as directional and the inference method is stated.

This is an independent third-party assessment. It is not a record of Reliance Jio Infocomm Ltd., is not prepared on the company's behalf, and is not affiliated with or endorsed by it. Prepared as an analytics capstone exercise.

Notes

1. Subscriber base, ARPU, traffic and financial indicators per company quarterly disclosures, Q1 FY26 through Q1 FY27.
2. Regulator monthly performance indicators; portability request volumes to November 2025.
3. Circle-level churn is not separately disclosed. Ranking inferred from portability request concentration; directional only.
4. Failure modes drawn from public consumer grievance channels. Indicative of failure type, not of incidence rate — no inference should be drawn about volume.
5. Regulator direction dated 27 February 2026 on AI/ML-based UCC detection and inter-operator intelligence sharing.